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LM-79-19 Test Report

For

American Lighting Industry Corp

(Brand Name: ASTRE LIGHTS)

7 Scouting Blvd,medford, NY 11763

Model name(s): ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1[Blank;(PIR);(PTC);(MS);(PIP);(MSP);(PR);(PSC);(BT)]

Report Type:	Testing and Report According to IES LM-79-2019
Type of Luminaire:	Outdoor Pole/Arm-Mounted Area and Roadway Luminaires
Report Date:	2025-04-28

Test & Report By:

Ferrum Li

Engineer: Ferrum Li

Review By:

Garman Mo

Manager: Garman Mo

Note: 1.The results contained in this report pertain only to the tested samples.
2.This report does not imply product certification, approval, or endorsement by A2LA or any agency of the Federal Government.

1.1 Product Information:		
Model Number	ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1[Blank;(PIR);(PTC);(MS);(PIP);(MSP);(PR);(PSC);(BT)]	
Remark	The "[Blank;(PIR);(PTC);(MS);(PIP);(MSP);(PR);(PSC);(BT)]" can be (PIR)=PIR sensor, (PTC)=Photocell; (MS)=Motion sensor; (PIP)=PIR sensor and Photocell, (MSP)=Motion sensor and Photocell; (PR)=PR Motion sensor; (PSC)=Photocell socket and shorting cap; (BT)=Bluetooth Motion sensor or blank=no sensor function.	
Representative (Tested) Model	ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1(0%,5000K) ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1(50%,5700K) ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1(100%,6500K)	
Model Difference	N/A	
SKU (if available)	--	
Type of Luminaire (for integral lamps, list base type and lamp type)	Outdoor Pole/Arm-Mounted Area and Roadway Luminaires	
LED Manufacturer	Bridgelux Inc.	
LED Model	BXEN-50E-13H-9D1 BXEN-65E-13H-9D1	
Integral Controls Availability	Yes	
Dimming	Continuous	
Sample Number	JDE250402-A1	
Date of Receipt	2025-04-11	
Luminaire Aperture (for downlights)	--	in.
Luminaire Length	--	mm
Luminaire Width	--	mm
Number of Units (modular products)	N/A	s

1.2 Rated Values:	
Rated Voltage / Frequency	120-277Vac, 50/60Hz
Nominal Power	70W/100W/150W (Power adjustable)
Rated Initial Lamp Lumen	--
Declared CCT	5000K,5700K,6500K(Color Tunable)

1.3 Test Specifications:

Test item	1. Total Luminous Flux 2. Luminous Distribution Intensity 3. Luminous Efficacy 4. Correlated Color Temperature 5. Color Rendering Index 6. Chromaticity Coordinate 7. Electrical Parameters
Reference Standard	1. IES LM-79-2019 Optical and Electrical Measurements of Solid-State Lighting Products 2. ANSI C78.377-2015 Specifications for the Chromaticity of Solid State Lighting Products 3. CIE 13.3-1995 Method of Measuring and Specifying Colour Rendering Properties of Light Sources 4. CIE 15-2004 Technical Report Colorimetry 5. IESNA LM-16-93 Practical Guide to Colorimetry of Light Source 6. IESNA TM-16-05 Technical Memorandum on Light Emitting Diode (LED) Sources and Systems
Reference Work Instruction	QD25

1.4 Test Methods

1) Photometric and Light Distribution Measurement – Goniophotometer Method: Photometric parameters were measured using the goniophotometer and software. The ambient temperature shall be maintained at 25 °C ±1.2 °C, measured at a point not more than 1.5 m from the sample and at the same height as the sample. The humidity should be maintained between 10% and 65%. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Luminous flux, luminaire efficacy, zonal lumen were calculated from the software taken at 1 °vertical intervals and 22.5 °horizontal intervals.
2) Chromaticity Measurement – Sphere-Spectroradiometer Method: Chromaticity parameters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 ºC ±1.2 ºC. The humidity should be maintained between 10% and 65%. The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral power distribution taken at 5 nm intervals over the range of 380 to 780 nm.
3) Electrical Measurements: Electrical parameters were measured using power meters incorporated in goniophotometer or sphere-spectroradiometer system. The ambient temperature surrounding the sample was maintained at 25 ºC ±1.2 ºC. The sample was operated at 120 or rated Volts AC, 60Hz. It was stabilized before measurement was made. Voltage, frequency, current, power, power factor and total harmonic distortion were measured by and read from the power meter.

2.1 Summary of Test Result

Criteria Item	Measured Value		Compliance	Requirement (DLC V5.1)	
Minimum Total Luminous	18561		Pass	>=1000(-10%)	
Minimum Luminous Efficacy	130.59		Pass	Standard: >= 105(-3%)	Premium: >= 120(-3%)
Minimum Power Factor	0.9099		Pass	>= 0.9(-3%)	
Maximum THD %	9.53		Pass	<= 20(+5)	
Minimum CRI	83.2		Pass	>= 70(-1)	
Minimum R9	11		Pass	>= 0(-1)	
Minimum Rg	94		Pass	>= 89(-1)	
Minimum Rf	83		Pass	>= 70(-1)	
Rcs, h1(%)	-13		Pass	-18%-23%(-1%)	
CCT (K)	5000K	5001	Pass	<=6500K	
	5700K	5741			
	6500K	6625			
Zonal Lumen Requirement(%)	0-90°	99.6	Pass	>=100(-1)	
	80-90°	2.9	Pass	<=10(+3)	
BUG rating	B3-U3-G3		--	--	

Note: Decision rule is the rule that describes how measurement uncertainty is accounted for when stating conformity with a specified requirement. When decision rule is applicable, please refer to the most updated version announced in our website:
https://en.bestway-compliance.com/pages/decision-rule-of-compliance-with-specification

2.2 Electrical, Photometric and Chromaticity Measurements

Test date	2025-04-14	Test Ambient:	25±1 ºC
Test Orientation	As intended	Stabilization Time (min)	60
Model Number	ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1(0%,5000K)	Total Operating Time (min)	75

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE250402-A1	120.0	60	1.202	143.6	0.9960	4.22
	277.0	60	0.5511	138.9	0.9099	9.53

Photometric Measurement – Goniophotometer Method(Test Distance:26.000m):

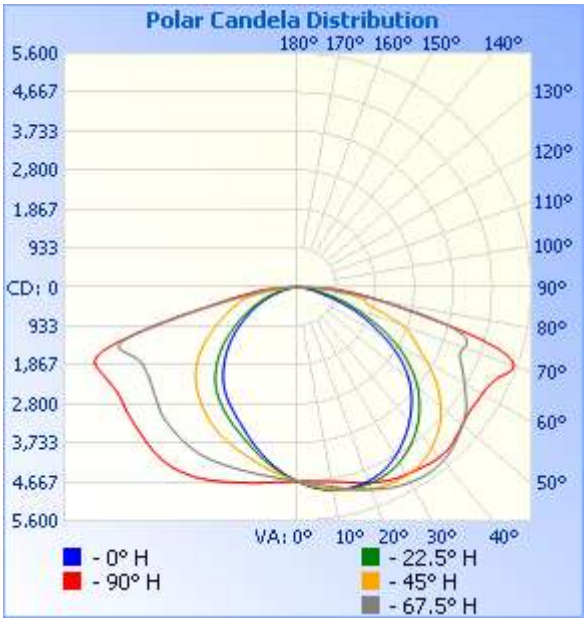
Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	18753	18561
Luminous Efficacy (lm/W)	130.59	133.63
Zonal lumens in the 0-90 °zone (%)	99.6	--
Zonal lumens in the 80-90 °zone (%)	2.9	--
BUG rating	B3-U3-G3	--
Beam Angle (°)	130.5	--
Center Beam Candle Power (cd)	4653	--

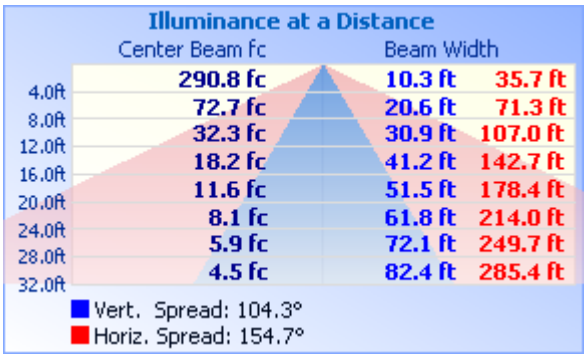
Zonal Lumen Tabulation

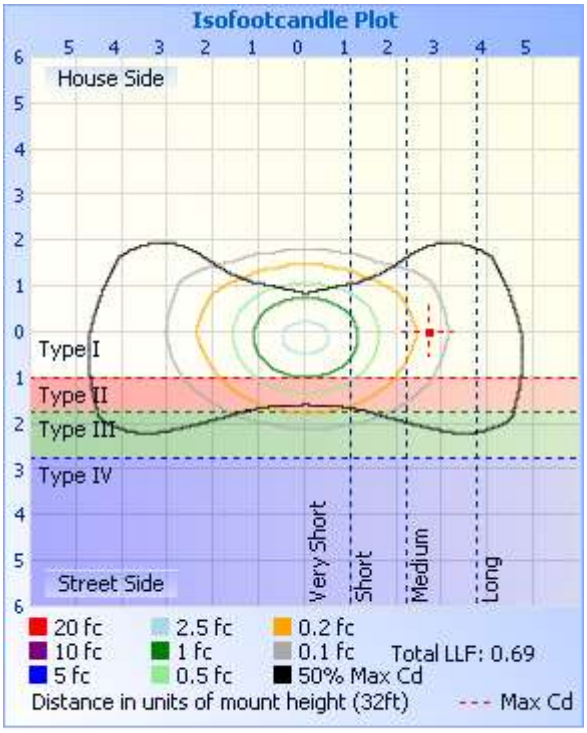
Zonal Lumen Summary		
Zone	Lumens	% Luminaire
0-30	3,904.3	20.8%
0-40	6,716.1	35.8%
0-60	13,130.9	70%
60-90	5,541.4	29.6%
70-100	2,642.8	14.1%
90-120	34.1	0.2%
0-90	18,672.3	99.6%
90-180	79.6	0.4%
0-180	18,751.9	100%

Lumens Per Zone					
Zone	Lumens	% Total	Zone	Lumens	% Total
0-10	444.3	2.4%	90-100	9.4	0.1%
10-20	1,318.8	7.0%	100-110	11.3	0.1%
20-30	2,141.2	11.4%	110-120	13.3	0.1%
30-40	2,811.8	15.0%	120-130	13.2	0.1%
40-50	3,203.7	17.1%	130-140	11.5	0.1%
50-60	3,211.1	17.1%	140-150	9.3	0%
60-70	2,908.0	15.5%	150-160	6.4	0%
70-80	2,094.7	11.2%	160-170	3.6	0%
80-90	538.7	2.9%	170-180	1.6	0%

Photometric Data







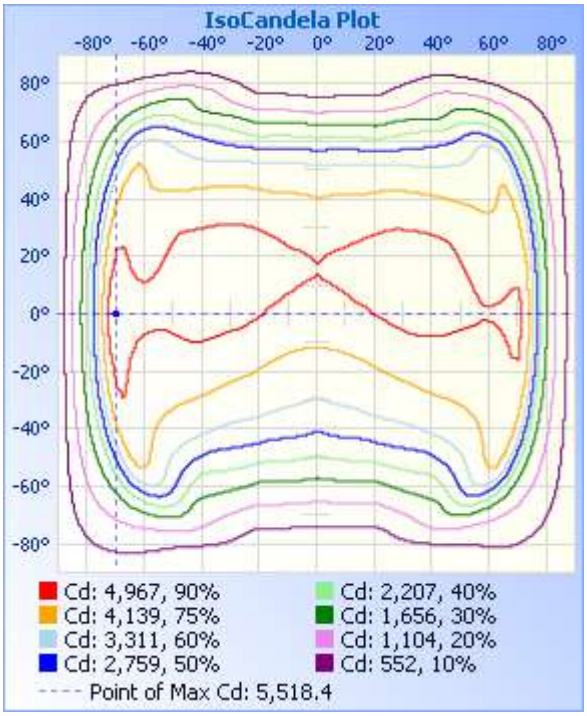
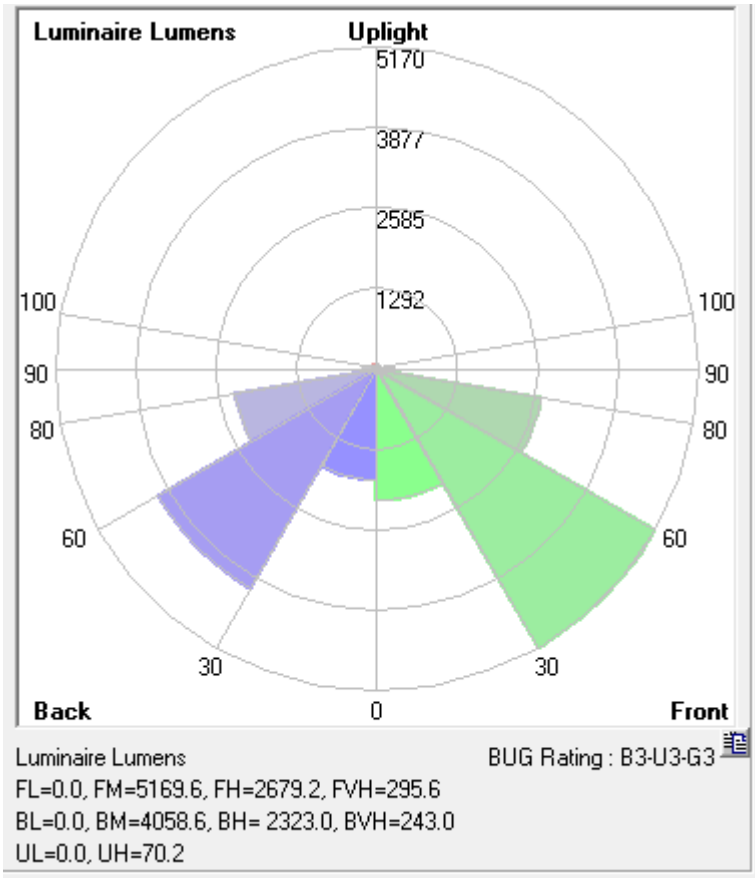


Table--1 UNIT: cd

γ (DEG) \ C (DEG)	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5			
0	4653	4653	4653	4653	4653	4653	4653	4653	4653	4653	4653	4653	4653	4653	4653	4653			
5	4685	4753	4798	4820	4824	4812	4791	4736	4668	4590	4509	4461	4441	4465	4522	4603			
10	4756	4875	4926	4939	4932	4934	4922	4855	4733	4563	4371	4253	4199	4256	4393	4579			
15	4848	5017	5040	5003	4970	4998	5037	5004	4843	4575	4253	4028	3935	4039	4267	4580			
20	4969	5166	5123	5020	4949	5013	5141	5158	4992	4627	4160	3817	3677	3828	4158	4596			
25	5092	5311	5172	4969	4855	4969	5207	5301	5113	4680	4076	3632	3449	3644	4056	4619			
30	5200	5414	5167	4834	4681	4857	5218	5402	5189	4665	3964	3465	3251	3469	3930	4639			
35	5262	5426	5090	4625	4447	4680	5152	5457	5254	4624	3810	3286	3040	3282	3778	4615			
40	5247	5353	4934	4393	4178	4439	5008	5429	5309	4584	3620	3057	2783	3061	3609	4558			
45	5180	5238	4685	4075	3867	4136	4793	5323	5266	4538	3419	2782	2473	2773	3424	4493			
50	5112	5025	4338	3696	3499	3766	4480	5159	5153	4428	3188	2454	2137	2425	3200	4377			
55	5052	4742	3889	3228	3014	3293	4085	4938	5088	4300	2879	2054	1781	2032	2890	4245			
60	5012	4385	3362	2636	2405	2736	3577	4626	5116	4206	2497	1647	1418	1615	2527	4161			
65	5095	3978	2919	1937	1784	2061	3118	4327	5239	4188	2078	1268	1086	1232	2136	4183			
70	5182	3931	2345	1357	1162	1441	2674	4243	5518	4448	1710	923	772	896	1766	4476			
75	3467	3672	1572	782	596	867	1862	3924	4130	3741	1398	577	454	549	1481	3594			
80	1620	1799	1288	370	277	420	1582	2205	2204	1748	1007	303	235	288	965	1373			
85	859	598	646	120	85.4	152	776	884	1199	875	326	122	86.1	107	263	737			
90	15.8	14.1	7.30	2.09	1.18	1.74	7.43	14.1	14.8	12.5	45.9	1.51	1.04	2.43	28.9	14.0			
95	16.6	13.8	6.50	1.90	0.97	1.35	6.55	13.5	14.1	11.9	6.13	2.55	1.73	3.12	7.78	14.2			
100	20.3	15.4	6.55	1.95	0.93	1.62	6.43	15.4	16.2	13.1	8.45	4.52	3.58	4.75	9.87	15.6			
105	23.1	16.7	7.16	2.44	1.62	2.43	7.32	16.4	18.5	15.6	11.0	6.02	5.32	6.36	11.9	17.9			
110	24.6	18.4	7.75	3.71	3.35	3.94	9.17	18.3	20.6	18.4	12.3	8.69	7.75	8.69	13.6	20.9			
115	25.6	19.0	9.25	4.05	3.93	4.19	10.4	20.1	21.8	20.2	13.9	8.23	10.1	9.44	14.7	21.7			
120	23.8	18.6	9.97	6.73	5.45	6.72	11.1	19.7	21.7	20.0	14.7	10.3	10.9	10.1	14.9	20.5			
125	21.8	17.9	9.71	8.64	7.75	8.46	11.2	18.7	18.6	19.5	14.6	14.5	16.4	13.8	15.1	20.0			
130	21.6	17.2	9.27	9.51	8.79	9.50	10.8	17.5	18.3	18.0	14.5	16.1	16.5	16.9	15.3	18.6			
135	20.9	16.4	9.00	10.4	9.18	10.7	10.3	15.8	18.3	18.2	14.4	16.8	16.2	18.1	15.1	18.8			
140	20.6	15.3	9.02	10.8	9.49	11.0	8.36	15.2	18.3	18.7	13.9	17.0	15.7	17.1	14.8	20.0			
145	20.6	14.4	10.6	11.0	10.7	12.2	9.77	14.9	18.3	18.1	13.9	16.3	14.5	16.9	14.9	20.0			
150	16.7	13.2	12.1	11.1	10.9	12.4	12.5	14.7	17.0	16.8	13.9	15.7	16.9	16.7	16.7	18.9			
155	14.3	11.6	12.7	11.2	11.1	12.2	13.3	12.9	14.7	14.6	14.0	15.0	14.9	16.1	16.4	17.6			
160	13.9	11.1	13.0	11.3	11.3	12.3	13.4	11.7	13.2	13.5	13.7	13.3	11.3	11.4	13.5	12.8			
165	12.5	11.4	13.2	11.0	10.9	11.8	13.5	11.8	13.3	13.8	13.5	12.2	11.0	11.9	11.6	12.1			
170	15.3	13.6	15.7	16.3	16.3	15.4	15.3	14.8	15.6	15.5	16.3	17.1	15.4	16.0	17.5	16.5			
175	16.6	16.2	17.2	16.4	18.0	16.6	16.8	16.8	18.0	18.1	16.3	17.3	16.8	19.8	16.7	18.1			
180	16.9	16.2	16.5	16.5	18.3	15.6	17.1	17.0	17.8	16.8	16.3	16.9	16.4	18.2	15.7	16.9			

BUG


Luminaire Lumens
Uplight
5170
3877
2585
1292
100
90
80
60
30
0
Back
Front
Luminaire Lumens
BUG Rating : B3-U3-G3
FL=0.0, FM=5169.6, FH=2679.2, FVH=295.6
BL=0.0, BM=4058.6, BH= 2323.0, BVH=243.0
UL=0.0, UH=70.2

2.3 Electrical, Photometric and Chromaticity Measurements

Test date	2025-04-14	Test Ambient:	25±1 ºC
Test Orientation	As intended	Stabilization Time (min)	60
Model Number	ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1(0%,5000K)	Total Operating Time (min)	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE250402-A1	120.0	60	1.209	144.6	0.9963	4.20
	277.0	60	0.5549	139.9	0.9102	9.51

Chromaticity Measurement - Sphere-Spectroradiometer Method(Self-absorption:1.1507) (4π geometry):

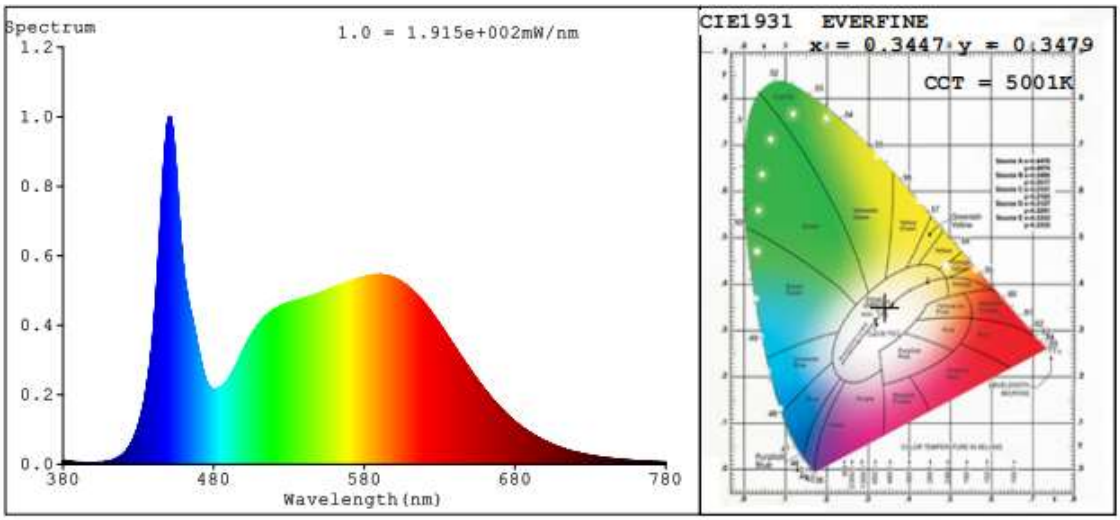
Parameter	Result
Test Voltage (V)	120
Frequency (Hz)	60
CCT (K)	5001
Duv	-0.0017
Chromaticity (x, y)	x=0.3447 y=0.3479
Chromaticity (u', v')	u'=0.2126 v'=0.4828

Parameter	Result
Color Rendering Index (CRI)	85.8
R9	22
Rg	98
Rf	84
Rcs,h1(%)	-11

Photometric Measurement –Sphere-Spectroradiometer Method:

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	18940	18746
Luminous Efficacy (lm/W)	130.98	134.00

Spectral Power Distribution & Chromaticity Diagram



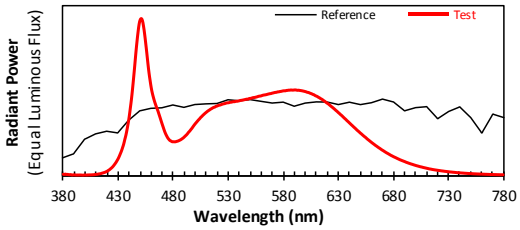
Special Color Rendering Indices

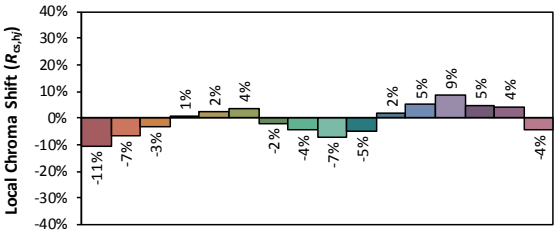
R1 =85	R2 =91	R3 =93	R4 =86	R5 =86	R6 =86	R7 =88	
R8 =72	R9 =22	R10=77	R11=86	R12=66	R13=87	R14=96	R15=81

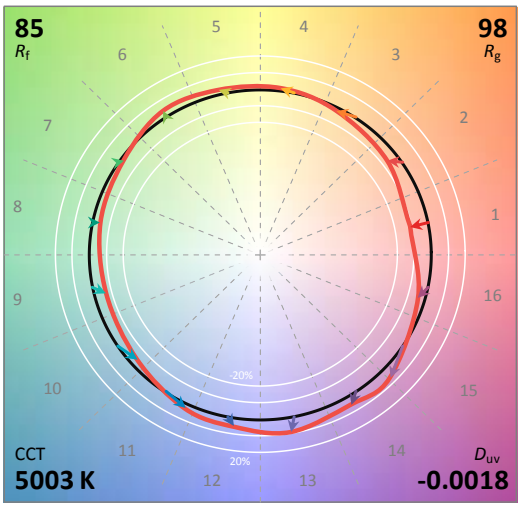
TM30

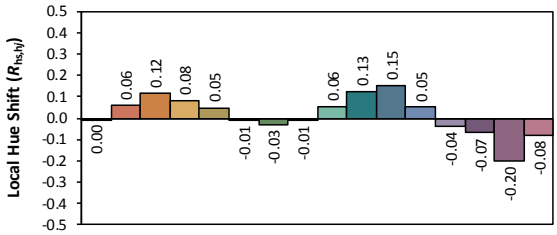
ANSI/IES TM-30-18 Color Rendition Report

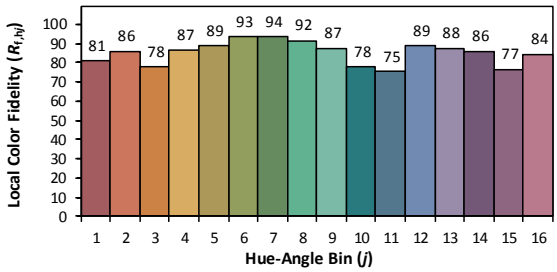
Source:	BXEN-50E-13H-9D1	Manufacturer:	American Lighting Industry Corp
Date:	2025-04-14	Model:	ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1 (0%, 5000K)

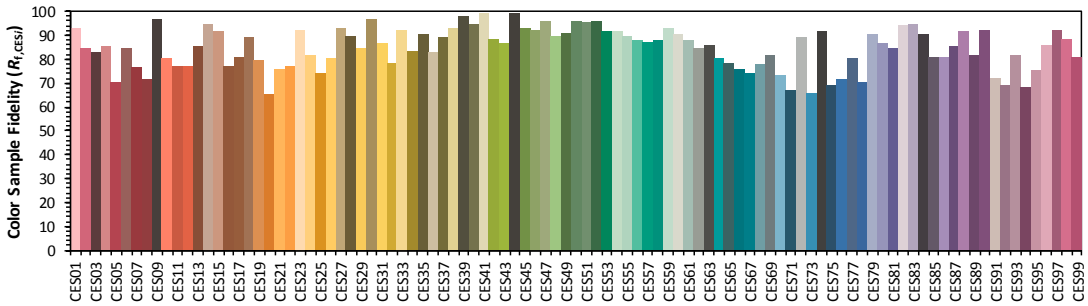












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3447
y	0.3477
u'	0.2126
v'	0.4827

CIE 13.3-1995 (CRI)

R_a	86
R_9	22

lors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.0

2.4 Electrical, Photometric and Chromaticity Measurements

Test date	2025-04-14	Test Ambient:	25±1 ºC
Test Orientation	As intended	Stabilization Time (min)	60
Model Number	ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1(50%,5700K)	Total Operating Time (min)	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE250402-A1	120.0	60	1.175	140.5	0.9965	4.18
	277.0	60	0.5389	135.9	0.9104	9.49

Chromaticity Measurement - Sphere-Spectroradiometer Method(Self-absorption:1.1510) (4π geometry):

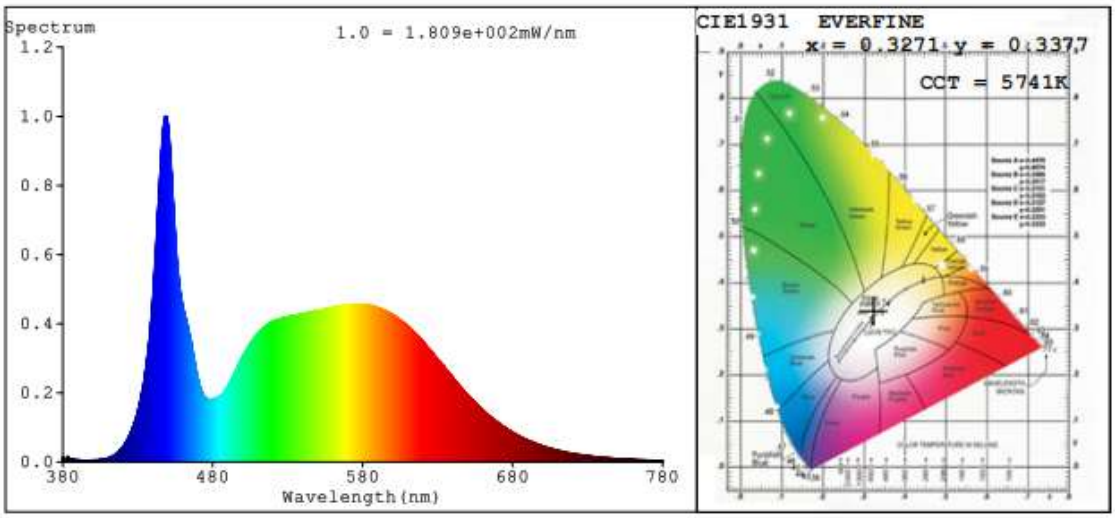
Parameter	Result
Test Voltage (V)	120
Frequency (Hz)	60
CCT (K)	5741
Duv	0.0007
Chromaticity (x, y)	x=0.3271 y=0.3377
Chromaticity (u', v')	u'=0.2045 v'=0.4750

Parameter	Result
Color Rendering Index (CRI)	83.2
R9	11
Rg	97
Rf	83
Rcs,h1(%)	-12

Photometric Measurement –Sphere-Spectroradiometer Method:

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	20557	20347
Luminous Efficacy (lm/W)	146.31	149.72

Spectral Power Distribution & Chromaticity Diagram



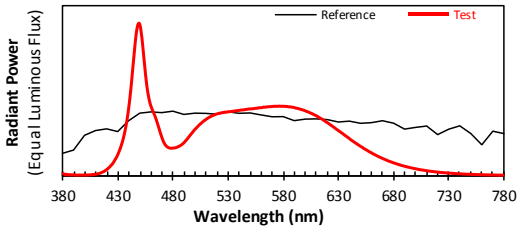
Special Color Rendering Indices

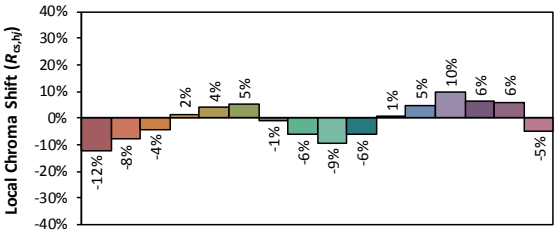
R1 =82	R2 =87	R3 =89	R4 =85	R5 =83	R6 =82	R7 =87	
R8 =70	R9 =11	R10=68	R11=85	R12=62	R13=83	R14=94	R15=78

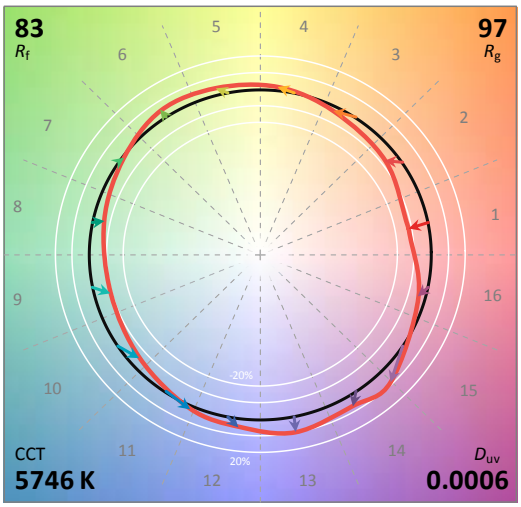
TM30

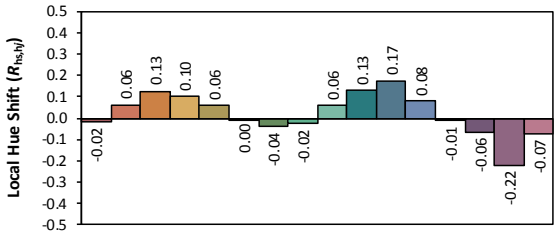
ANSI/IES TM-30-18 Color Rendition Report

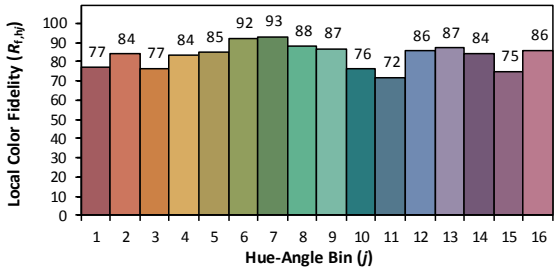
Source:	BXEN-50E-13H-9D1 BXEN-65E-13H-9D1	Manufacturer:	American Lighting Industry Corp
Date:	2025-04-14	Model:	ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1(50%, 5700K)

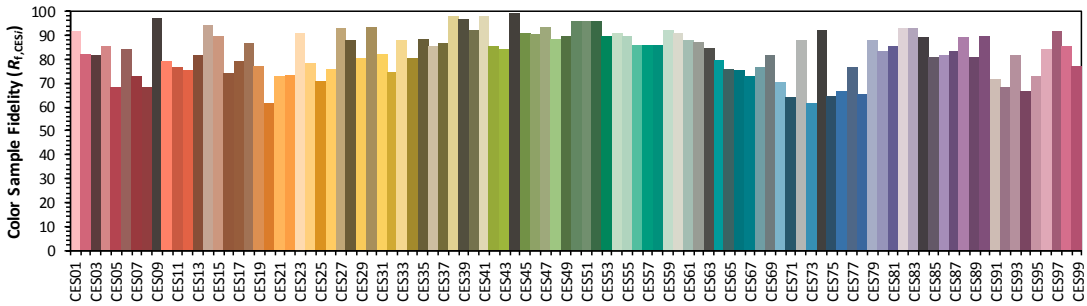












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3271
y	0.3375
u'	0.2045
v'	0.4749

CIE 13.3-1995 (CRI)	
R_a	83
R_9	11

lors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.0

2.5 Electrical, Photometric and Chromaticity Measurements

Test date	2025-04-14	Test Ambient:	25±1 ºC
Test Orientation	As intended	Stabilization Time (min)	60
Model Number	ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1(100%,6500K)	Total Operating Time (min)	61

Electrical Measurement:

Sample No.	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	THD %
JDE250402-A1	120.0	60	1.189	142.2	0.9966	4.17
	277.0	60	0.5452	137.5	0.9104	9.48

Chromaticity Measurement - Sphere-Spectroradiometer Method(Self-absorption:1.1508) (4π geometry):

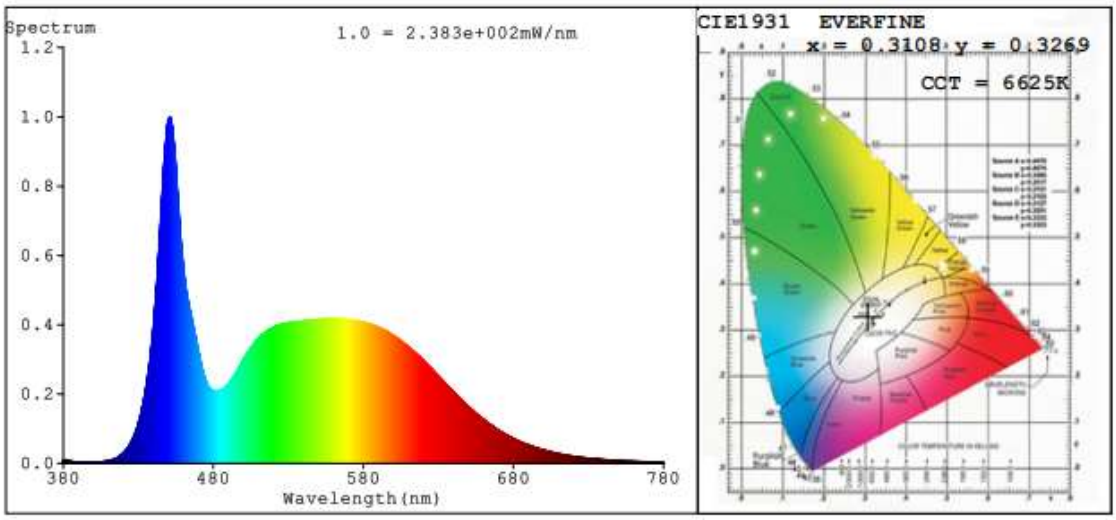
Parameter	Result
Test Voltage (V)	120
Frequency (Hz)	60
CCT (K)	6625
Duv	0.0030
Chromaticity (x, y)	x=0.3108 y=0.3269
Chromaticity (u', v')	u'=0.1973 v'=0.4669

Parameter	Result
Color Rendering Index (CRI)	83.8
R9	11
Rg	94
Rf	84
Rcs,h1(%)	-13

Photometric Measurement –Sphere-Spectroradiometer Method:

Parameter	Result	
Test Voltage (V)	120	277
Frequency (Hz)	60	60
Total Luminous (lm)	19826	19623
Luminous Efficacy (lm/W)	139.42	142.71

Spectral Power Distribution & Chromaticity Diagram



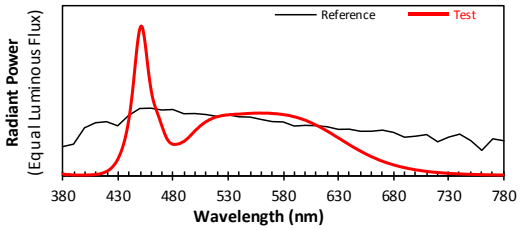
Special Color Rendering Indices

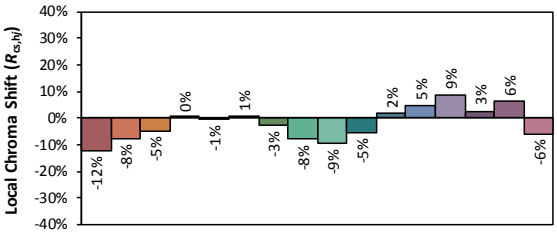
R1 =82	R2 =88	R3 =91	R4 =84	R5 =83	R6 =83	R7 =89	
R8 =71	R9 =11	R10=71	R11=83	R12=57	R13=84	R14=95	R15=78

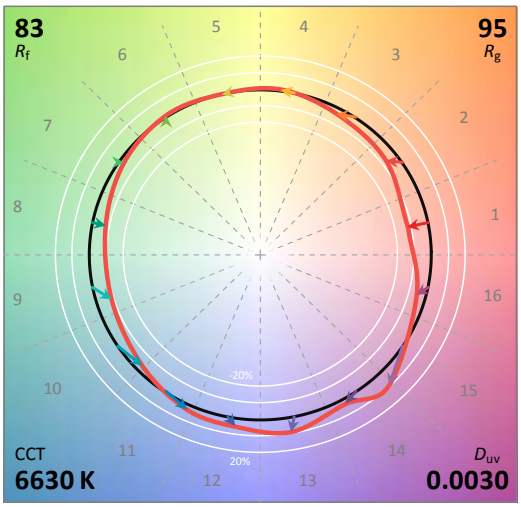
TM30

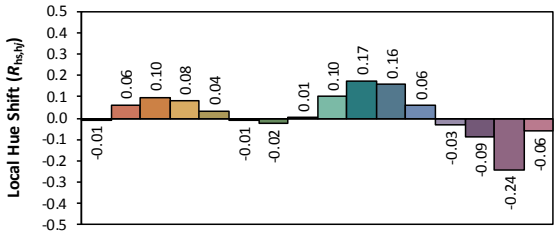
ANSI/IES TM-30-18 Color Rendition Report

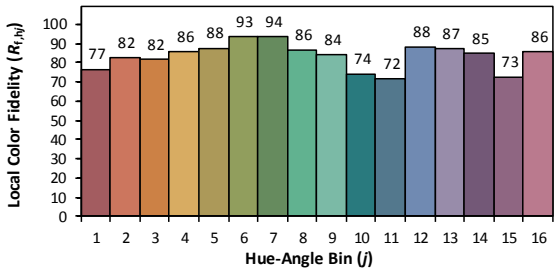
Source:	BXEN-65E-13H-9D1	Manufacturer:	American Lighting Industry Corp
Date:	2025-04-14	Model:	ALI-SBX-70/100/150W-5000/5700/6500K-H3-S1(100%, 6500K)

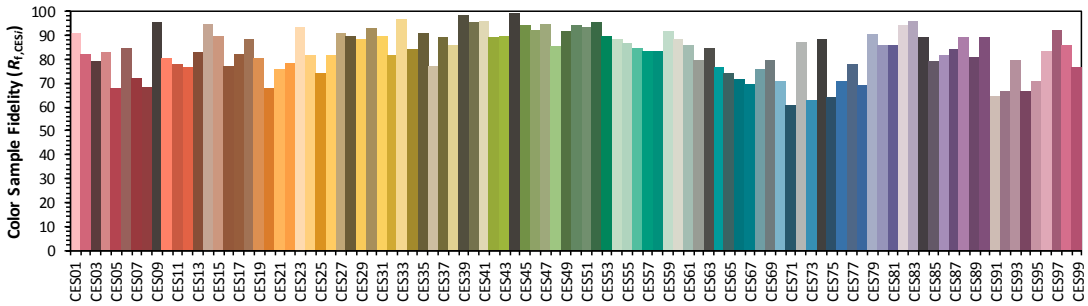












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3108
y	0.3267
u'	0.1974
v'	0.4668

CIE 13.3-1995 (CRI)	
R_a	84
R_9	12

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3. Test Equipment

Equipment ID	Equipment Name	Last Calibration Date	Next Calibration Date
ST-R-S-451	2 meter Integrating Sphere	Verified by D204 standard lamp	
ST-R-S-327	Spectral analysis system HAAS-2000	Verified by D204 standard lamp	
ST-R-S-452	Standard Lamp D204	2023-06-26	2026-06-25
ST-R-S-453	Power Meter for Integrating Sphere	2024-05-29	2025-05-28
ST-R-S-405	Hygrothermograph	2024-06-06	2025-06-05
ST-R-355	Goniophotometer system	Verified by D908S standard lamp	
ST-R-359	Standard Lamp D908S	2022-07-19	2025-07-18
ST-R-S-357	AC Power Source	2024-12-23	2025-12-22
ST-R-S-422	Power Meter for Goniophotometer	2024-05-29	2025-05-28
ST-R-S-354	hygrothermograph for Goniophotometer	2024-05-29	2025-05-28
Uncertainty: Photometric Measurement (Sphere):2.94%, k=2 Chromaticity Measurement(Sphere):45.06K, k=2 Photometric Measurement(Goniophotometer):3.32%, k=2			

4. Product Photo

******** END OF REPORT ********